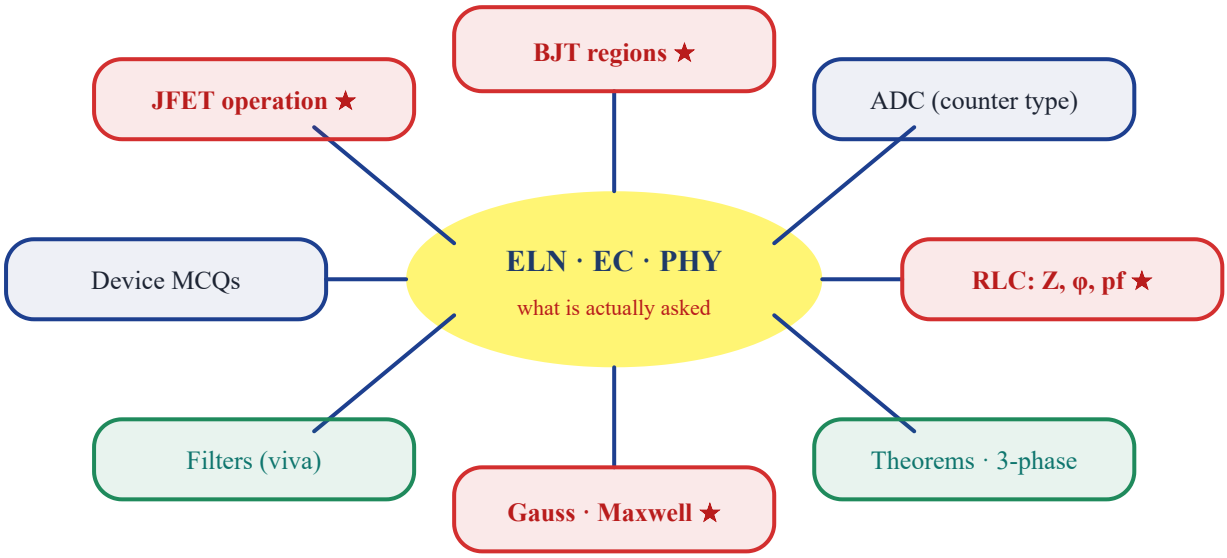
Solved Past Questions: Electronics, Circuits & Physics

22 questions from BCS, BPSC, bank and BUET-set papers, solved in 10 sheets



↑ the asked topics on one page: red = asked most often, green = viva / one-mark

Papers covered

- **49th BCS** ICT (281), MCQ paper, 2025: 12 MCQs
- **45th BCS** Written Technical Cadre (ICT-281), 31.01.2024: Q27(a), (b)
- **Bangladesh Bank** AME (BIBM), 04.02.2023: Q10 · **Petrobangla** AM (IT) (BUET), 16.09.2022: Q6, Q8
- **BPSC viva 2023** (Asst. Maintenance Engineer) · two written items and one MCQ from recruitment papers whose name is not printed

Most asked

① n-JFET for V_{GS}, V_{DS}
② BJT cut-off, active, saturation
③ series RLC: $|Z|$ and φ
④ Gauss's law + Maxwell
⑤ max power: $R_L = R_{Th}$
⑥ $\beta = \alpha/(1-\alpha)$

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example · blue frame = the question as asked

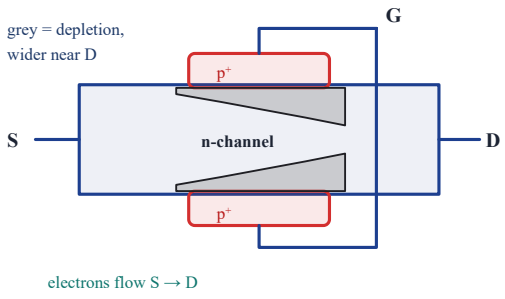
Contents

✓ JFET: working principle, regions …… p. 2
✓ BJT regions, load line, α–β …… p. 3
✓ Counter-type ADC …… p. 4
✓ Devices MCQs + filters viva …… p. 5
✓ Circuits: solved past questions …… p. 6
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✓ Physics: Maxwell, waves, quantum …… p. 8
✓ Night-before sheet …… p. 9
✓ Practice MCQs + written …… p. 10

1 n-Channel JFET: Working Principle & Regions

Question: Explain the working principle of an n-channel JFET for various values of V_{GS} and V_{DS}.

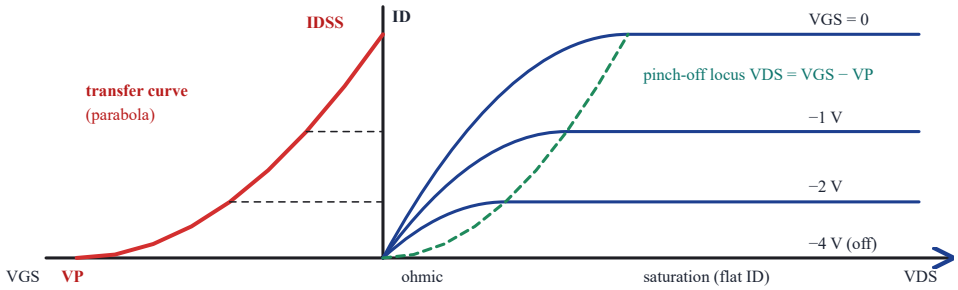
Bangladesh Bank AME (BIBM) · 04.02.2023 · Q10(b) · also asked elsewhere with "operating regions"



p^+ gates tied together

A JFET is a **voltage-controlled, unipolar** device. The gate is a **reverse-biased p–n junction**, so $I_G \approx 0$ and $R_{in} \approx 10^8$–$10^{10}\ \Omega$.

1. **$V_{GS} = 0$, small V_{DS}:** channel fully open; I_D rises **linearly** (ohmic region, a resistor)
2. **$V_{GS} = 0$, V_{DS} rising:** reverse bias is largest at the drain end, so the depletion layers grow wedge-shaped. At $V_{DS} = |V_P|$ they meet: **pinch-off**; I_D levels at $\mathbf{I_{DSS}}$
3. **V_{GS} negative:** channel narrower everywhere; pinch-off at $V_{DS} = V_{GS} - V_P$, lower saturated I_D
4. **$V_{GS} \le V_{GS(off)} = V_P$:** channel closed, $I_D \approx 0$: **cut-off**
5. **Very large V_{DS}:** avalanche breakdown at the drain end



left: I_D vs V_{GS} (transfer) · right: drain characteristics, $I_{DSS} = 10$ mA, $V_P = -4$ V, to scale

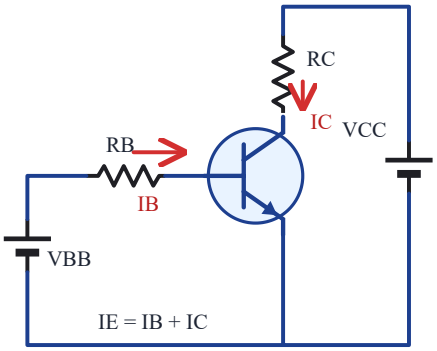
Region ($V_P < 0$)	Condition	Behaviour
Cut-off	$V_{GS} \le V_P$	$I_D \approx 0$, open switch
Ohmic (triode)	$V_{DS} < V_{GS} - V_P$	voltage-controlled R
Saturation (pinch-off)	$V_{DS} \ge V_{GS} - V_P$	$\mathbf{I_D = I_{DSS}(1 - V_{GS}/V_P)^2}$

Memory trick: JFET = a **garden hose**: V_{GS} squeezes it, V_{DS} pinches the drain end; squeeze fully (V_P) and flow stops. Normally **ON**.

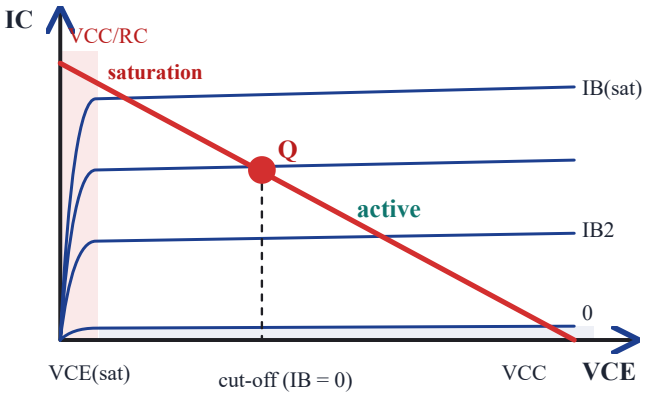
Exam trap: A JFET gate is **not insulated** (that is the MOSFET) and a JFET has **no V_{th}**: it conducts at $V_{GS} = 0$ and cuts off at $V_{GS(off)} = V_P$.

2 BJT: Cut-off, Active & Saturation Regions

Question: Describe the cut-off, saturation and active regions of operation of a transistor with a diagram.
Bangladesh Bank AME (BIBM) · 04.02.2023 · Q10(a) · also asked in a recruitment paper with the CE load line



common-emitter (CE) npn circuit



I_C–V_{CE} curves with the dc load line (red)

1. **Load line** (KVL): $\mathbf{V_{CE} = V_{CC} - I_C R_C}$, from $(V_{CC}, 0)$ to $(0, V_{CC}/R_C)$; Q slides along it as I_B changes
2. **Cut-off:** line meets the $I_B = 0$ curve; only leakage I_{CEO} flows, $V_{CE} \approx V_{CC}$: an **open switch**
3. **Saturation:** line meets $I_{B(sat)}$; $\mathbf{I_{C(sat)} \approx V_{CC}/R_C}$, $V_{CE} = V_{knee} \approx 0.2$ V; more I_B cannot raise I_C: a **closed switch**
4. **Active:** in between; $I_C = \beta I_B$, the transistor **amplifies** (Q mid-way for full swing)

Region	B–E	C–B	npn values	Use
Cut-off	reverse	reverse	$I_B = 0$, $I_C = I_{CEO}$	switch OFF
Active	**forward**	**reverse**	$V_{BE} \approx 0.7$ V, $I_C = \beta I_B$	amplifier
Saturation	forward	forward	$V_{CE} \approx 0.2$ V, $I_C < \beta I_B$	switch ON

Worked example ✍️: $V_{CC} = 12$ V, $R_C = 2$ kΩ, $\beta = 100$, $I_B = 43$ µA → $I_C = 4.3$ mA, $V_{CE} = 3.4$ V: **active**. $I_B = 215$ µA → $\beta I_B = 21.5$ mA > $I_{C(sat)} = 11.8/2k = 5.9$ mA: **saturated**.

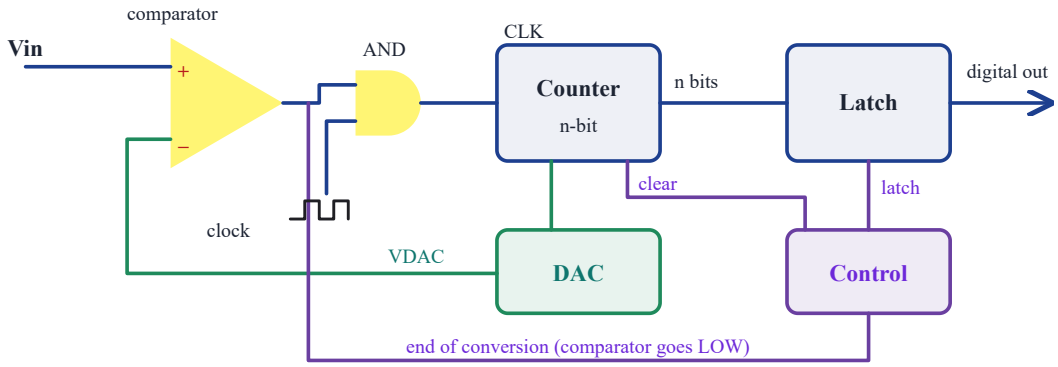
α and β (MCQ): $I_E = I_B + I_C$; $\alpha = I_C/I_E$, $\beta = I_C/I_B = I_C/(I_E - I_C)$ → $\mathbf{\beta = \alpha/(1 - \alpha)}$, $\alpha = \beta/(1 + \beta)$.
Asked with options $\beta = \alpha/(1+\alpha)$, $1 - \alpha$, $\alpha = \beta^2$, **$\alpha/(1 - \alpha)$** → (d). $\alpha = 0.98 \rightarrow \beta = 49$.

Exam trap: In saturation $I_C = \beta I_B$ is **false**: the circuit caps I_C at $\approx V_{CC}/R_C$. BJT "saturation" (switch ON) is **not** FET "saturation" (the flat, amplifying region).

3 Counter-Type (Digital-Ramp) ADC

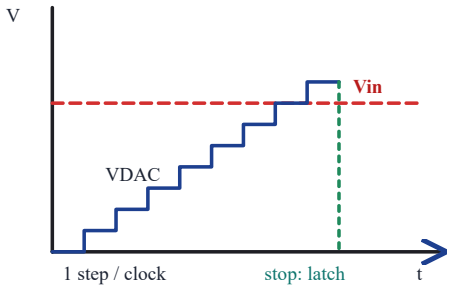
Question: Draw an ADC circuit which converts an analog signal to a digital signal, and explain it.
Petrobangla Assistant Manager (IT) (BUET) · 16.09.2022 · Q6

An ADC turns a voltage V_{in} into an n-bit number. The **counter-type ADC** counts clock pulses, turns the count back into a voltage with a **DAC**, and stops the moment that voltage passes V_{in}.



counter-type ADC: comparator, AND gate, counter, DAC in the feedback path, latch and control

1. **Start:** counter cleared, $V_{DAC} = 0 < V_{in}$, comparator **HIGH**
2. HIGH opens the **AND gate**: clock pulses reach the counter
3. DAC output rises as a **staircase**, 1 LSB per clock
4. $V_{DAC} > V_{in}$ → comparator **LOW** → clock blocked
5. Control **latches** the count (the output) and clears the counter



the staircase climbs until it passes V_{in}

Worked example ✍️: Step = $\mathbf{V_{ref}/2^n}$. 8-bit, $V_{ref} = 5.12$ V, 1 MHz clock, $V_{in} = 3.01$ V: step = 20 mV; counting stops at the first N with $0.02N > 3.01$ → N = **151** = 1001 0111 ($V_{DAC} = 3.02$ V), after 151 µs. Worst case $(2^n - 1)$ clocks = 255 µs.

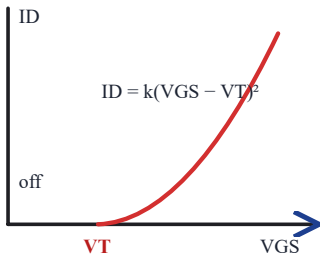
Other ADCs: flash = 1 clock, $2^n - 1$ comparators (fastest) · SAR = n clocks, fixed (most MCUs) · dual-slope = slowest, rejects noise (multimeters). Sample at $f_s \geq 2f_{max}$ (Nyquist).

Memory trick: Counter ADC = "**climb the stairs until your head hits the ceiling**" (V_{in}); SAR = **binary search**, MSB first.

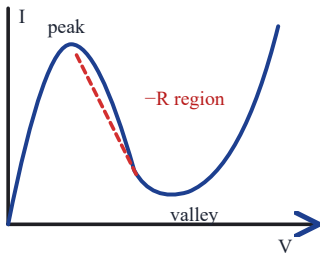
Exam trap: The counter ADC's conversion time is **not constant**: it grows with V_{in}. A fixed n-clock time belongs to SAR.

4 Electronic Devices: MCQs Asked

Asked	Answer	Why
Active component? (resistor / transistor / inductor / capacitor)	**Transistor**	gives power gain; R, L, C are passive
Gate voltage for an n-channel E-MOSFET? ($+V_{DD}$ / $-V_{DD}$ / 0)	**$+V_{DD}$**	$V_{GS} > V_T$ induces the channel
Max η, transformer-coupled class A? (25 / 30 / 50 / 78 %)	**50%**	no dc loss in a load resistor
Working principle of a tunnel diode?	**Quantum tunnelling**	heavy doping → very thin junction



n-E-MOSFET transfer curve



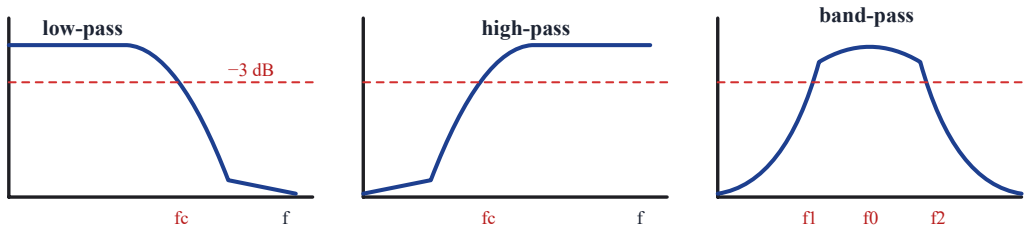
tunnel diode I–V

Class (angle)	η max
A, R load (360°)	25%
A, transformer	**50%**
B (180°)	78.5%
C (< 180°)	> 90%

Why 50%: $P_{dc} = V_{CC}I_{CQ}$; the transformer lets V_{CE} swing $0 \rightarrow 2V_{CC}$, so $P_{ac(max)} = (V_{CC}/\sqrt{2})(I_{CQ}/\sqrt{2}) = V_{CC}I_{CQ}/2 \rightarrow$ **$\eta = 50\%$**. Class B: $\pi/4 = 78.5\%$.

5 Filters

Viva: Can you explain low-pass and band-pass signals / filters? (Asst. Maintenance Engineer)

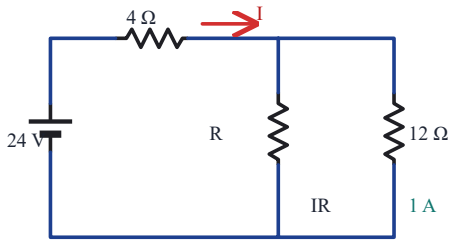


pass band above the dashed −3 dB (0.707) line

- **Low-pass** (series R, shunt C): passes $f < f_c$; **$f_c = 1/(2\pi RC)$**. **High-pass** (series C, shunt R): $f > f_c$
- −3 dB = half power = 0.707 V_{max}; 1st order rolls off 20 dB/decade
- **Band-pass:** f_1–f_2; **BW = $f_2 - f_1$**, $f_0 = \sqrt{(f_1 f_2)}$, Q = f_0/BW
- Low-pass signal = baseband (speech); band-pass signal = around a carrier (AM/FM)

6 Circuits: Solved Past Questions

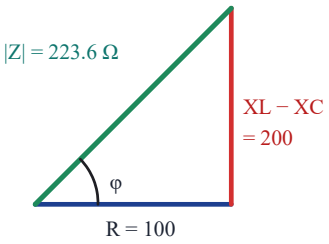
Question: Find R and I from the given circuit. Petrobangla AM (IT) (BUET) · 16.09.2022 · Q8 (figure not reproduced; the standard type)



given: 1 A flows in the 12 Ω branch

1. Ohm: V across the parallel pair = 1 × 12 = **12 V**
2. KVL: 4 Ω drops 24 − 12 = 12 V → **I = 12/4 = 3 A**
3. KCL: I_R = 3 − 1 = 2 A → **R = 12/2 = 6 Ω**
4. Check: 4 + (6 ∥ 12) = 8 Ω; 24/8 = 3 A ✓

MCQ asked (49th BCS ICT, 2025)	Answer	Why
Superposition rests on?	**Linearity**	add single-source responses
Max power transfer when?	**$R_L = R_S$**	$dP/dR_L = 0$ there
pf of a purely resistive load?	**1**	φ = 0, cos 0° = 1
Series RLC impedance?	**$\sqrt{[R^2 + (X_L - X_C)^2]}$**	X_L, X_C 180° apart
Balanced star: V_L vs V_P?	**$V_L = \sqrt{3}\, V_P$**	phasors 120° apart
R 100, X_L 300, X_C 100 Ω: phase?	**63.43° (nearest 60°)**	$\tan^{-1}(200/100)$



impedance triangle

$|Z| = \sqrt{(100^2 + 200^2)}$ = **223.6 Ω** · φ = $\tan^{-1}(2)$ = **63.43°** · pf = cos φ = R/|Z| = **0.447 lagging** (inductive, V leads I).

Exam trap: $\tan^{-1}(2)$ = 63.43°, **not** 60° ($\tan 60° = 1.732$). No option is exact: 60° is only the nearest.

Max power transfer: $P = V_{Th}^2 R_L/(R_{Th} + R_L)^2$ → max at $R_L = R_{Th}$: **$P_{max} = V_{Th}^2/4R_{Th}$**, η = 50%. AC: $Z_L = Z_{Th}^*$ (conjugate). 12 V, 4 Ω → 9 W.

Three-phase: Star: $V_L = \sqrt{3}V_P$, $I_L = I_P$ · Delta: $V_L = V_P$, $I_L = \sqrt{3}I_P$ · $P = \sqrt{3}V_L I_L \cos\varphi$. 230 V phase → 398 ≈ **400 V** line.

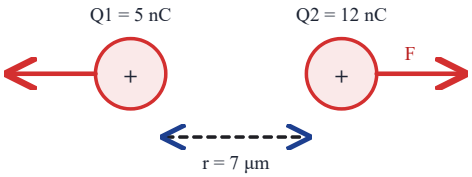
Memory trick: superposition: "**One at a time, the rest asleep**": V-sources shorted, I-sources opened; add currents, **never powers** ($P \propto I^2$).

7 Physics: E vs B Field · Coulomb's Law

Question: Mention the differences between electric field and magnetic field. Two charges $Q_1 = 5$ nC and $Q_2 = 12$ nC are 7 μm apart in vacuum. Find the force exerted on Q_2 by Q_1.

45th BCS Written Technical Cadre (ICT-281) · 31.01.2024 · Q27(a)

Point	Electric field E	Magnetic field B
Source	any charge (at rest too)	moving charges, magnets, changing E
Acts on	every charge: **F = qE**	only **moving** charges: **F = q(v × B)**
Work	can do work on q	does **no work** (F ⊥ v)
Field lines	start on +, end on −	**closed loops** (∇·B = 0)
Unit	N/C = V/m	tesla (T) = Wb/m²



like charges repel

Solution ✍️: $F = kQ_1Q_2/r^2$, $k = 1/(4\pi\varepsilon_0) = 9 \times 10^9$ N·m²/C²

$= 9 \times 10^9 \times (5 \times 10^{-9})(12 \times 10^{-9}) / (7 \times 10^{-6})^2$

$= (9 \times 10^9 \times 6 \times 10^{-17}) / (4.9 \times 10^{-11})$

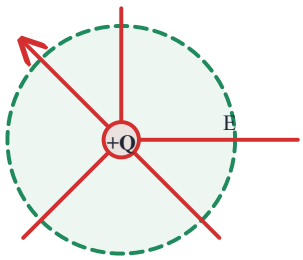
$= 5.4 \times 10^{-7} / 4.9 \times 10^{-11} =$ **1.10×10^4 N**, **repulsive**, along the line from Q_1 to Q_2.

Exam trap: Convert first: nC = 10^{-9} C, μm = 10^{-6} m → $r^2 = 4.9 \times 10^{-11}$ m². ≈ 11 kN is right (only 7 μm apart). The numerical carries marks: don't stop at the differences.

8 Gauss's Law

Question: State and explain Gauss's law. 45th BCS · 31.01.2024 · Q27(b)

Statement: the total electric flux through any **closed surface** = **net enclosed charge ÷ ε_0**: $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{enc}/\varepsilon_0$ ($\oint D \cdot dA = Q_{enc}$).



Gaussian sphere, radius r

flux depends only on Q inside

- Charges **outside** the surface add zero net flux (their lines enter and leave)
- Shape of the surface does not matter: pick one where E is constant and ⊥
- **Point charge:** $E(4\pi r^2) = Q/\varepsilon_0 \rightarrow E = Q/(4\pi\varepsilon_0 r^2)$: Coulomb's law follows
- Line λ: $E = \lambda/(2\pi\varepsilon_0 r)$ · sheet σ: $E = \sigma/2\varepsilon_0$

9 Maxwell's Equations

Question: Write Maxwell's equations in differential and integral forms for (i) static EM fields, (ii) time-varying EM fields. 45th BCS · 31.01.2024 · Q27(b)

Law	Differential form	Integral form
Gauss (E)	$\nabla \cdot D = \rho_v$	$\oint D \cdot dS = Q_{enc} = \int \rho_v \, dv$
Gauss (B)	$\nabla \cdot B = 0$	$\oint B \cdot dS = 0$
Faraday	$\nabla \times E = -\partial \mathbf{B}/\partial t$	$\oint E \cdot dl = -(d/dt) \int \mathbf{B} \cdot \mathbf{dS}$
Ampère–Maxwell	$\nabla \times H = J + \partial \mathbf{D}/\partial t$	$\oint H \cdot dl = \int (J + \partial \mathbf{D}/\partial t) \cdot dS$

(i) **Static fields:** drop the highlighted terms → $\nabla \times E = 0$, $\oint E \cdot dl = 0$, $\nabla \times H = J$, $\oint H \cdot dl = I_{enc}$ · (ii) **Time-varying:** all four as written

- Only the two **curl** laws change; $\partial D/\partial t$ = displacement current
- Media: $\mathbf{D = \varepsilon E}$, $\mathbf{B = \mu H}$, $J = \sigma E$
- Free space ($\rho = 0$, $J = 0$): the equations give a wave at $\mathbf{c = 1/\sqrt{(\mu_0 \varepsilon_0)} \approx 3 \times 10^8}$ **m/s**

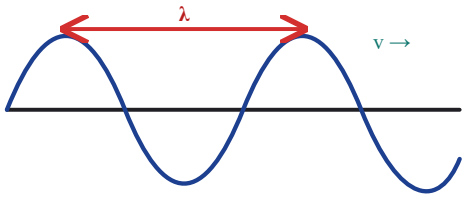
Memory trick: **"Two divs, two curls"**: div D = charge, div B = nothing; curl E = minus B-dot, curl H = J plus D-dot. Integral form = the same law via the divergence (div) and Stokes (curl) theorems.

Exam trap: $\nabla \times E = 0$ holds **only for static** fields. In the time-varying set, missing $\partial D/\partial t$ (Maxwell's own addition, 1865) loses the marks.

10 Waves & Quantum: MCQs Asked

Q. Relation between velocity v, frequency f and wavelength λ? (v = f/λ / v = λ/f / **v = fλ** / v = f + λ) → **v = fλ**: in one period T = 1/f the wave moves one λ.

FM 100 MHz → $\lambda = 3 \times 10^8/10^8$ = **3 m** · Wi-Fi 2.4 GHz → **12.5 cm** · sound 340 m/s at 1 kHz → 34 cm.



one wavelength per period

Q. Which equation describes the wave nature of particles? (**Schrödinger** / Maxwell / Newton's wave / Gauss–Jordan) → **Schrödinger**. Maxwell's equations describe EM waves; Gauss–Jordan is a matrix method.

Quantum one-liners: Time-independent: $\mathbf{-(\hbar^2/2m)\nabla^2\psi + V\psi = E\psi}$ · time-dependent: $i\hbar \, \partial\psi/\partial t = \hat{H}\psi$ (Schrödinger, 1926) · $|\psi|^2$ = probability density · de Broglie $\mathbf{\lambda = h/p = h/mv}$ · photon $E = hf$ · $h = 6.63 \times 10^{-34}$ J·s

Night-Before-the-Exam Sheet

Formula sheet:

JFET: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$

JFET saturates at $V_{DS} \geq V_{GS} - V_P$

E-MOSFET: $I_D = k(V_{GS} - V_T)^2$, $V_{GS} > V_T$

BJT: $I_E = I_B + I_C$, $I_C = \beta I_B$

$\beta = \alpha/(1 - \alpha) \cdot \alpha = \beta/(1 + \beta)$

Load line $V_{CE} = V_{CC} - I_C R_C$

ADC step $= V_{ref}/2^n$

Counter ADC $\leq 2^n - 1$ clocks · SAR $= n$

$f_c = 1/(2\pi RC) \cdot BW = f_2 - f_1$

$|Z| = \sqrt{(R^2 + (X_L - X_C)^2)}$

$\varphi = \tan^{-1}((X_L - X_C)/R) \cdot pf = R/|Z|$

$P_{max} = V_{Th}^2/4R_{Th}$ at $R_L = R_{Th}$

Star $V_L = \sqrt{3}V_P$ · delta $I_L = \sqrt{3}I_P$

$F = kq_1q_2/r^2$, $k = 9 \times 10^9$

$\oint E \cdot dA = Q_{enc}/\varepsilon_0$

$\nabla \times E = -\partial B/\partial t \cdot \nabla \times H = J + \partial D/\partial t$

$v = f\lambda \cdot c = 1/\sqrt{(\mu_0\varepsilon_0)}$

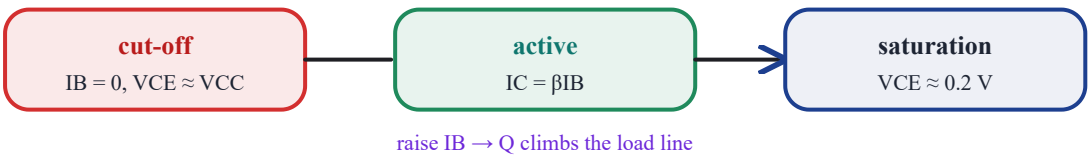
$\lambda = h/p \cdot E = hf$

One-liners for the MCQ paper

1. Active: transistor; R, L, C passive
2. n-E-MOSFET: gate +, $V_{GS} > V_T$
3. Class A: 50% (transformer), 25% (R)
4. Class B $\eta = 78.5\%$; class C > 90%
5. Tunnel diode: tunnelling, −R region
6. JFET: normally ON, p–n gate
7. BJT active: B–E forward, C–B reverse
8. Superposition needs linearity
9. Max power: $R_L = R_S$, efficiency 50%
10. pf: R load = 1; pure L or C = 0
11. R 100, X_L 300, X_C 100 → $\varphi = 63.43°$
12. Wave nature of particles: Schrödinger
13. $v = f\lambda$; −3 dB = half power = 0.707 V
14. Magnetic force does no work

Top traps: JFET has no V_{th} and no insulated gate (that is the MOSFET) · BJT saturation ≠ FET saturation · in BJT saturation $I_C < \beta I_B$ · $\tan^{-1}2 = 63.43°$, not 60° · superpose currents, never powers · $\nabla \times E = 0$ only for static fields · convert nC and μm before Coulomb

All tricks together: JFET = garden hose · junctions F-R = gain, F-F = ON, R-R = OFF · counter ADC climbs stairs, SAR does binary search · superposition: one at a time, the rest asleep · Maxwell: two divs, two curls



the three BJT regions in one line

✓ Practice MCQs (exam pattern, 10 marks)

1. If $\alpha = 0.99$, β is:
(a) 0.99 (b) 99
(c) 100 (d) 9.9

2. n-JFET, $V_P = -4$ V, $V_{GS} = -1$ V: least V_{DS} to saturate:
(a) 5 V (b) 1 V
(c) 3 V (d) 4 V

3. Max η of class-B push-pull:
(a) 78.5% (b) 50%
(c) 25% (d) 100%

4. An n-channel E-MOSFET conducts when:
(a) $V_{GS} = 0$ (b) $V_{GS} < 0$
(c) $V_{GS} < V_T$ (d) $V_{GS} > V_T$

5. 8-bit counter ADC, 1 MHz: worst-case time:
(a) 8 µs (b) 255 µs
(c) 128 µs (d) 1 ms

6. Series R = 30, $X_L = 90$, $X_C = 50$ Ω: $|Z|$ =
(a) 170 Ω (b) 70 Ω
(c) 50 Ω (d) 40 Ω

7. $V_{Th} = 10$ V, $R_{Th} = 5$ Ω: P_{max} =
(a) 20 W (b) 10 W
(c) 2.5 W (d) 5 W

8. Star, $V_L = 400$ V: $V_P \approx$
(a) 231 V (b) 400 V
(c) 693 V (d) 133 V

9. λ of a 50 MHz wave in vacuum:
(a) 0.6 m (b) 6 m
(c) 60 m (d) 1.5 m

10. Flux through a closed surface, no charge inside:
(a) Q/ε_0 (b) infinite
(c) zero (d) shape-dependent

Answer key → 1-b · 2-c · 3-a · 4-d · 5-b · 6-c · 7-d · 8-a · 9-b · 10-c

Written question: An n-channel JFET has $I_{DSS} = 10$ mA and $V_P = -5$ V. (a) Find I_D at $V_{GS} = -2$ V. (b) Find the minimum V_{DS} for saturation. (c) Find g_m there. (d) Name the region if $V_{GS} = -6$ V. 8 marks

1. $I_D = 10(1 - (-2)/(-5))^2 = 10(0.6)^2 = \mathbf{3.6\ mA}$ 3
2. $V_{DS(min)} = V_{GS} - V_P = -2 + 5 = \mathbf{3\ V}$ 2
3. $g_{m0} = 2I_{DSS}/|V_P| = 4$ mS; $g_m = 4(1 - 0.4) = \mathbf{2.4\ mS}$ 2
4. $V_{GS} = -6$ V $< V_P \rightarrow$ **cut-off**, $I_D \approx 0$ 1

Before the exam ✓

- ✓ Can sketch JFET drain and transfer curves
- ✓ Can draw the CE load line, 3 regions
- ✓ Can draw the counter ADC and its staircase
- ✓ Can write all 8 Maxwell forms from memory

Also practise (written):
① Derive $\beta = \alpha/(1 - \alpha)$
② SAR ADC for $V_{in} = 3.01$ V
③ Gauss's law → E of a line charge
④ Max power: prove $R_L = R_{Th}$